

COA-MODULE-2- part-2 Continued

SIGNED OPERAND MULTIPLICATION

BOOTH ALGORITHM

- This algorithm
 - generates a $2n$ -bit product
 - treats both positive & negative 2's-complement n -bit operands uniformly (Figure 9.9-9.12).
- Attractive feature: This algorithm achieves some efficiency in the number of addition required when the multiplier has a few large blocks of 1s.
- This algorithm suggests that we can reduce the number of operations required for multiplication by representing multiplier as a difference between 2 numbers.
For e.g. multiplier(Q) 14(001110) can be represented as

$$\begin{array}{r} 010000 \text{ (16)} \\ -000010 \text{ (2)} \\ \hline 001110 \text{ (14)} \end{array}$$
- Therefore, product $P=M*Q$ can be computed by adding 2^4 times the M to the 2's complement of 2^1 times the M.

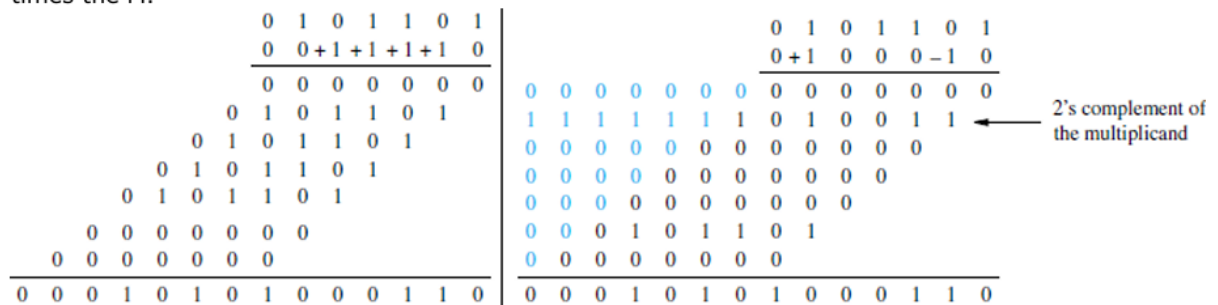


Figure 9.9 Normal and Booth multiplication schemes.

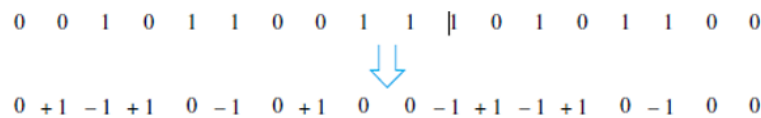


Figure 9.10 Booth recoding of a multiplier.

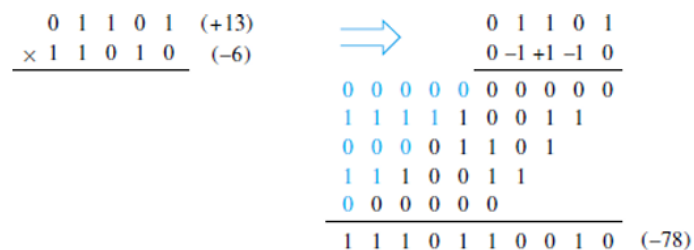


Figure 9.11 Booth multiplication with a negative multiplier.

Multiplier		Version of multiplicand selected by bit i
Bit i	Bit $i-1$	
0	0	$0 \times M$
0	1	$+1 \times M$
1	0	$-1 \times M$
1	1	$0 \times M$

Figure 9.12 Booth multiplier recoding table.

Booth Algorithm

Multiplier		Version of multiplicand selected by bit i
Bit i	Bit $i-1$	
0	0	0 X M
0	1	+ 1 X M
1	0	- 1 X M
1	1	0 X M

Booth multiplier recoding table.

Booth Algorithm

- Best case – a long string of 1's (skipping over 1s)
- Worst case – 0's and 1's are alternating

Worst-case multiplier	0	1	0	1	0	1	0	1	0	1	0	1	0	1
	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1

Ordinary multiplier	1	1	0	0	0	1	0	1	1	0	1	1	1	0	0
	0	-1	0	0	+1	-1	+1	0	-1	+1	0	0	0	-1	0

Good multiplier	0	0	0	0	1	1	1	1	1	0	0	0	0	1	1	1
	0	0	0	+1	0	0	0	0	-1	0	0	0	+1	0	0	-1

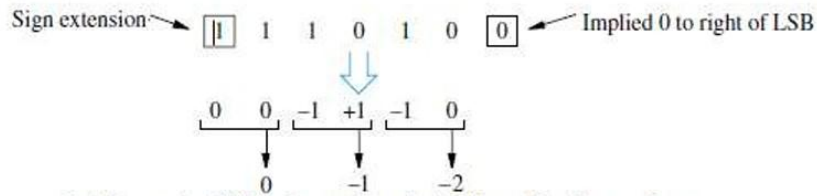
COA-MODULE-2- part-2 Continued

COMPUTER ORGANIZATION

FAST MULTIPLICATION

BIT-PAIR RECODING OF MULTIPLIERS

- This method
 - derived from the booth algorithm
 - reduces the number of summands by a factor of 2
- Group the Booth-recoded multiplier bits in pairs. (Figure 9.14 & 9.15).
- The pair (+1 -1) is equivalent to the pair (0 +1).



(a) Example of bit-pair recoding derived from Booth recoding

Multiplier bit-pair		Multiplier bit on the right	Multiplicand selected at position i
$i+1$	i	$i-1$	
0	0	0	$0 \times M$
0	0	1	$+1 \times M$
0	1	0	$+1 \times M$
0	1	1	$+2 \times M$
1	0	0	$-2 \times M$
1	0	1	$-1 \times M$
1	1	0	$-1 \times M$
1	1	1	$0 \times M$

(b) Table of multiplicand selection decisions

Figure 9.14 Multiplier bit-pair recoding.

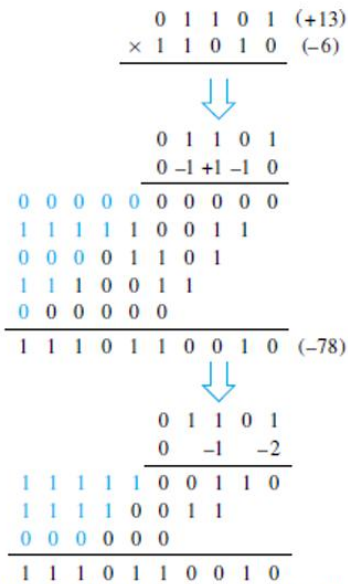


Figure 9.15 Multiplication requiring only $n/2$ summands.

COA-MODULE-2- part-2 Continued

COMPUTER ORGANIZATION

INTEGER DIVISION

- An n -bit positive-divisor is loaded into register M.
An n -bit positive-dividend is loaded into register Q at the start of the operation.
Register A is set to 0 (Figure 9.21).
- After division operation, the n -bit quotient is in register Q, and the remainder is in register A.

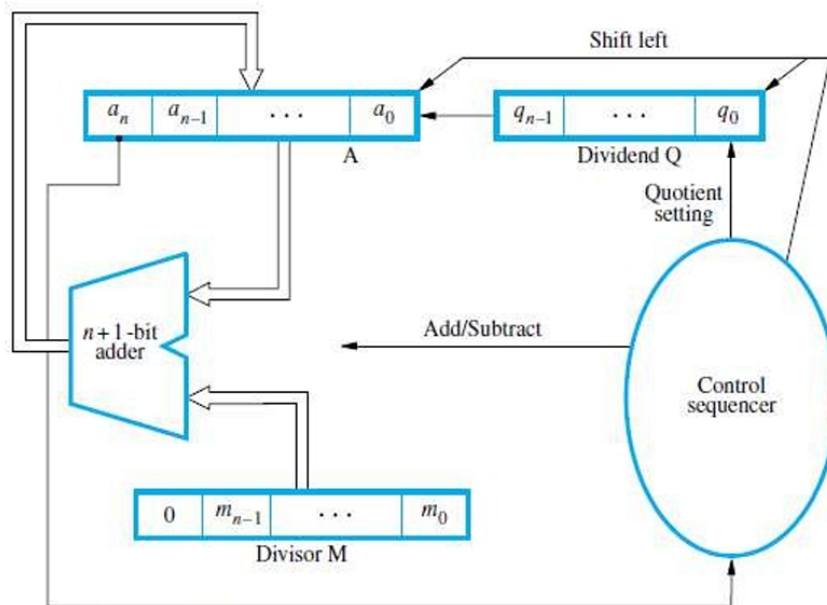


Figure 9.23 Circuit arrangement for binary division.

$$\begin{array}{r} 21 \\ 13 \overline{) 274} \\ \underline{26} \\ 14 \\ \underline{13} \\ 1 \end{array} \qquad \begin{array}{r} 10101 \\ 1101 \overline{) 100010010} \\ \underline{1101} \\ 10000 \\ \underline{1101} \\ 1110 \\ \underline{1101} \\ 1 \end{array}$$

Figure 9.22 Longhand division examples.

COA-MODULE-2- part-2 Continued

COMPUTER ORGANIZATION

RESTORING DIVISION

- Procedure: Do the following n times

- 1) Shift A and Q left one binary position (Figure 9.22).
- 2) Subtract M from A, and place the answer back in A
- 3) If the sign of A is 1, set q_0 to 0 and add M back to A (restore A).
If the sign of A is 0, set q_0 to 1 and no restoring done.

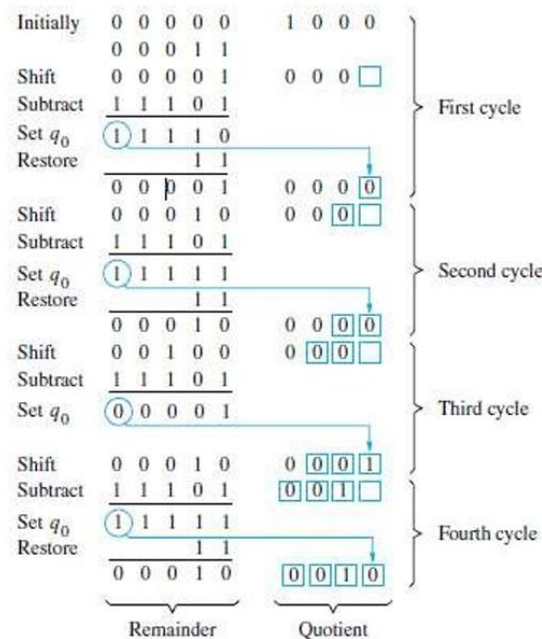


Figure 9.24 A restoring division example.

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COMPUTER ORGANIZATION

NON-RESTORING DIVISION

• Procedure:

Step 1: Do the following n times

- i) If the sign of A is 0, shift A and Q left one bit position and subtract M from A; otherwise, shift A and Q left and add M to A (Figure 9.23).
- ii) Now, if the sign of A is 0, set q_0 to 1; otherwise set q_0 to 0.

Step 2: If the sign of A is 1, add M to A (restore).

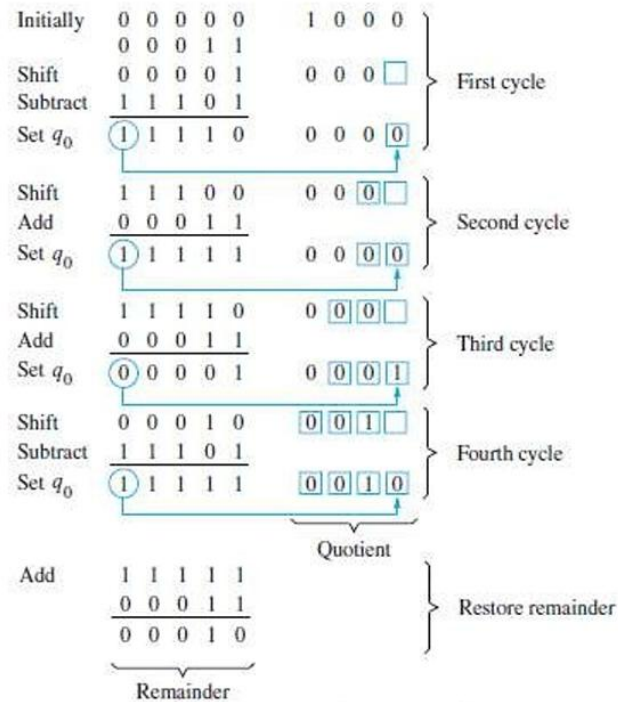


Figure 9.25 A non-restoring division example.

COA-MODULE-2- part-2 Continued

COMPUTER ORGANIZATION

Problem 4:

Perform signed multiplication of following 2's complement numbers using Booth's algorithm.

(a) A=010111 and B=110110 (b) A=110011 and B=101100

(c) A=110101 and B=011011 (d) A=001111 and B=001111

(e) A=10100 and B=10101 (f) A=01110 and B=11000

Solution:

$$\begin{array}{r}
 \begin{array}{r}
 010111 \\
 \times 110110 \\
 \hline
 \end{array}
 \quad
 \begin{array}{r}
 +23 \\
 \times -10 \\
 \hline
 -230
 \end{array}
 \end{array}$$

sign extension

$$\begin{array}{r}
 \begin{array}{r}
 111111 \\
 000000 \\
 111111 \\
 \hline
 111100011010
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r}
 110011 \\
 \times 101100 \\
 \hline
 \end{array}
 \quad
 \begin{array}{r}
 -13 \\
 \times -20 \\
 \hline
 260
 \end{array}
 \end{array}$$

sign extension

$$\begin{array}{r}
 \begin{array}{r}
 110011 \\
 000000 \\
 110011 \\
 \hline
 000100000100
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r}
 110101 \\
 \times 011011 \\
 \hline
 \end{array}
 \quad
 \begin{array}{r}
 -11 \\
 \times 27 \\
 \hline
 -297
 \end{array}
 \end{array}$$

sign extension

$$\begin{array}{r}
 \begin{array}{r}
 110101 \\
 000000 \\
 110101 \\
 \hline
 111011010111
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r}
 001111 \\
 \times 001111 \\
 \hline
 \end{array}
 \quad
 \begin{array}{r}
 15 \\
 \times 15 \\
 \hline
 225
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r}
 001111 \\
 000000 \\
 001111 \\
 \hline
 000011100001
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r}
 10100(-12) \\
 \times 10101(-11) \\
 \hline
 \end{array}
 \rightarrow
 \begin{array}{r}
 \begin{array}{r}
 10100 \\
 -11111 \\
 \hline
 01000001100 \\
 1111001000 \\
 00011000 \\
 1101000 \\
 011000 \\
 \hline
 0100000100(+132)
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{r}
 01110(+14) \\
 \times 11000(-8) \\
 \hline
 \end{array}
 \rightarrow
 \begin{array}{r}
 \begin{array}{r}
 01110 \\
 0-1000 \\
 \hline
 0000000000 \\
 0000000000 \\
 0000000000 \\
 11100100 \\
 000000 \\
 \hline
 1110010000(-112)
 \end{array}
 \end{array}$$

COA-MODULE-2- part-2 Continued

COMPUTER ORGANIZATION

Problem 5:

Perform signed multiplication of following 2's complement numbers using bit-pair recoding method.

(a) A=010111 and B=110110

(b) A=110011 and B=101100

(c) A=110101 and B=011011

(d) A=001111 and B=001111

Solution:

$$\begin{array}{r}
 010111 \\
 \times 110110 \\
 \hline
 \end{array}
 \qquad
 \begin{array}{r}
 010111 \\
 -1 \quad +2 \quad -2 \\
 \hline
 111111 \quad 1010010 \\
 00000101110 \\
 1111021011 \\
 \hline
 111100011010
 \end{array}$$

$$\begin{array}{r}
 110011 \\
 \times 101100 \\
 \hline
 \end{array}
 \qquad
 \begin{array}{r}
 110011 \\
 -1 \quad -1 \quad 0 \\
 \hline
 00000001101 \\
 0000011101 \\
 \hline
 000100000100
 \end{array}$$

$$\begin{array}{r}
 110101 \\
 \times 011011 \\
 \hline
 \end{array}
 \qquad
 \begin{array}{r}
 110101 \\
 +2 \quad -1 \quad -1 \\
 \hline
 000000001011 \\
 00000001011 \\
 11101011 \\
 \hline
 111011010111
 \end{array}$$

$$\begin{array}{r}
 001111 \\
 \times 001111 \\
 \hline
 \end{array}
 \qquad
 \begin{array}{r}
 001111 \\
 +1 \quad -1 \\
 \hline
 1111111110001 \\
 00001111 \\
 \hline
 000011100001
 \end{array}$$

COA-MODULE-2- part-2 Continued

COMPUTER ORGANIZATION

Problem 6:

Given A=10101 and B=00100, perform A/B using restoring division algorithm.

Solution:

Initially	000000 (A)	10101 (Q)
	000100 (M)	
Shift	000001	0101
Subtract	111100	
Set q0	111101	
Restore	100	
Shift	000001	01010
Subtract	000010	1010
Set q0	111101	
Restore	100	
Shift	000010	10100
Subtract	000101	0100
Set q0	111101	
No restore	000001	
No restore	000000	
	000001	10100
Shift	000001	01001
Subtract	111100	
Set q0	111101	
Restore	100	
Shift	000010	10010
Subtract	000101	0010
Set q0	111101	
No restore	000001	
No restore	000000	
	000001	00101
remainder		quotient

Problem 7:

Given A=10101 and B=00101, perform A/B using non-restoring division algorithm.

Solution:

	000000	10101	
	A	Q	
	000101		Initial configuration
	M		
shift	000001	0101	
subtract	111011		1st cycle
	111100	01010	
shift	111000	1010	
add	000101		2nd cycle
	111101	10100	
shift	111011	0100	
add	000101		3rd cycle
	000000	01001	
shift	000000	1001	
subtract	111011		4th cycle
	111011	10010	
shift	110111	0010	
add	000101		5th cycle
	111100	00100	
add	000101		
	000001		quotient
			remainder